

Directed Graph Based Carbon Flow Tracing for Demand Side Carbon Obligation Allocation

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Abstract—In order to achieve carbon emission abatement, some researchers and policy makers have cast their focus on demand side carbon abatement potentials. This paper addresses the problem of carbon flow calculation in power systems and carbon obligation allocation at demand side. A directed graph based method for tracing carbon flow is proposed. In a lossy network, matrices such as carbon losses, net carbon intensity (NCI) and footprint carbon intensity (FCI) are obtained with the proposed method and used to allocate carbon obligation at demand side. Case studies based on realistic distribution and transmission systems are provided to demonstrate the effectiveness of the proposed method.

Index Terms—carbon flow, carbon obligation, demand response, graph theory, low-carbon electricity

I. INTRODUCTION

Power industry, as one of the major industries consuming fossil fuel, attracts the interest of researchers towards carbon emission abatement [1]. In general, relevant policies focus on carbon obligation allocation in generation side, such as feed-in tariffs, carbon taxes, and cap-and-trade. However, the carbon obligation is finally transferred through complicated marketing processes to the consumers in terms of electricity bills, which may result in price volatility and speculations [2]. Therefore, it is worthwhile to also investigate the possibility of directly allocating carbon obligation at demand side.

In some regions, the obligation to reduce carbon emission is placed on electricity consumers, such as the UK Carbon Reduction Commitment (CRC) Energy Efficiency Scheme, the Tokyo Emission Trading Scheme and the Perform Achieve and Trade (PAT) Scheme in India [3]. A load-based cap-and-trade scheme has also been proposed in several states of the United

States as a regulatory approach to reduce carbon emission from the electric power industry [4]. Therefore, it is important to know how each load will be assigned carbon obligation.

A direct method to allocate this obligation is the proportional method based on electricity consumption, but the network characteristics and locational features are ignored. The concept of marginal carbon intensity (MCI) is proposed in [5] assessing the impact of locational carbon emission activities in constrained power networks through sensitivity analysis. In [6, 7], the concept of "carbon emission flow in networks" is introduced to illustrate the relationship between carbon emission and energy flow, and the corresponding calculation method is established. A modified carbon emission flow model based on life cycle assessment considering power losses is established in [8]. In [9], a linear equation-based carbon flow tracing method is proposed to address the carbon obligation allocation issue from a consumption-based perspective. However, a linear equation-based method requires a matrix inverse calculation which is time consuming especially for a large scale power system.

In this paper, a directed graph based method is proposed to calculate the carbon flow in lossy power systems. The method is a complement for the existing tracing methods, and can obtain all the key elements of the carbon flow such as the carbon loss, the net carbon intensity (NCI) and the footprint carbon intensity (FCI), etc. The NCI can be used to help measure carbon losses, and the FCI can be used to distinguish the location with low-carbon potential thereby guiding the carbon reducing activities. Both of them help address carbon obligation allocation at demand side.

II. CARBON FLOW THEORY REVISIT

Carbon flow is defined as a virtual network flow attached to power flow passing through power networks. By adding "carbon label" to power flows, the spatial movement of carbon emission in networks is demonstrated. Tracing power flow is a problem of determining the contributions of individual generators to loads and flows [10]. So is the carbon flow tracing. Based on the proportional sharing principle, power flow and

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the corresponding carbon flow passing through branches and nodes can be traced and calculated. Since the output and the carbon emission rate of generators are known, the upstream-looking algorithm [11] can be used to trace the carbon flow.

When considering network losses, the power flows at the beginning and the end of each branch are not the same. In order to get the upstream distribution matrix, some modifications such as average line flows, gross flows, net flows [9, 11], are used to generate a lossless network from the lossy one.

The equations of power flow tracing can then be denoted by $\mathbf{A}_u \mathbf{P}_N = \mathbf{P}_G$ and $\mathbf{P}_N = \mathbf{A}_u^{-1} \mathbf{P}_G$, where $\mathbf{P}_G, \mathbf{P}_N \in R^N$ are the vectors of generations and through-flows of nodes, and $\mathbf{A}_u$ is the upstream distribution matrix.

The elements of $\mathbf{P}_N$ can be expressed as:

$$P_{Ni} = \sum_{k=1}^N [A_u^{-1}]_{ik} P_{Gk} \quad (i = 1, 2, \dots, N). \quad (1)$$

The average carbon emission rate of generators at node k is represented by σ_k . Then the carbon flow passing through a particular node can be expressed as:

$$CF_{Ni} = \sum_{k=1}^N [A_u^{-1}]_{ik} P_{Gk} \sigma_k \quad (i = 1, 2, \dots, N). \quad (2)$$

The carbon intensity of a node, which is equivalent to the associated emission of unit active power at the node in generation side, is defined as:

$$CI_i = CF_{Ni} / P_{Ni} \quad (i = 1, 2, 3, \dots, N) \quad (3)$$

The carbon flow of branch ij is denoted as:

$$CF_{ij} = CI_i \times |P_{ij}| \quad (i \in \Gamma_-(j)) \quad (4)$$

where P_{ij} is the active power flow in branch ij and $\Gamma_-(j)$ is the set of injecting nodes of node j .

III. DIRECTED GRAPH BASED CARBON FLOW TRACING

In this section a directed graph based carbon flow tracing method is proposed. Several assumptions have to be made first:

- The flow of electricity follows the proportional sharing rule.
- The generator has the priority to supply the load on the same bus, which divides buses into pure generators and pure loads [12].

Power networks do not need to be transformed to equivalent lossless networks but are left as their lossy natures for further processing. In the directed graph of a lossy network, if the two terminals of a line have different active power flow directions (both terminals inject power into the line, often seen in a large scale power system), the line flow is treated as open circuit. Since losses of these lines are omitted, the allocated carbon obligation cannot fulfill the total emission. One treatment for this is to regard the two injections as equivalent loads on the two terminal buses respectively. The carbon obligation of the equivalent loads should be placed on the existing loads at the terminal buses, while on the network operator if no loads exist at the terminals.

A. Directed graph based carbon flow tracing method

The directed graph based carbon flow tracing method is carried out in a downstream way. We start from a pure generation node (which has net output into the network). Its outflows bring generators' power (carbon flow attached) toward the downstream nodes. Then delete this node with its incident lines to make a new graph which is a sub-graph of the previous one. Find the new pure generation node in the sub-graph and repeat the process described above until all lines have been deleted in

the graph. This process is shown in detail in Fig. 1, where K_i is the number of generators at node i , P_{i-k} and σ_{i-k} is the output and emission rate of the k th generator at node i , CF_{Ni}^{net} is the net nodal through-carbon-flow at node i and CF_{Ni}^{gross} is the gross nodal through-carbon-flow at node i . According to the usage of power flows in lossy networks, the tracing process includes tracing for NCIs and FCIs. The two carbon intensities are to be further analyzed as we discuss carbon obligation allocation.

B. Contribution factors

Though carbon flow indices can be derived directly from the process of eliminating nodes and branches in the directed graph, the contribution factors of generators to nodal through-carbon-flows can also be calculated during the tracing process as the linear equation-based method does but without matrix inverse calculations.

The element of contribution factor matrix D is defined as

$$CF_{Ni}^{gross} = \sum_{k=1}^N D_{ik} P_{Gk} \sigma_k \quad (i = 1, 2, \dots, N). \quad (5)$$

where D_{ik} can be calculated during the tracing process as follow (take FCI tracing process for example):

$$D_{ik} = \begin{cases} 1 & (k = i) \\ \sum_{m \in \Gamma_-(i)} (|P_{mi}| / P_{Nm} \cdot D_{mk}) & (k \neq i, \exists m \in \Gamma_-(i), D_{mk} > 0) \\ 0 & (\text{otherwise}) \end{cases} \quad (6)$$

It can be proved that the proposed method can uniquely determine the NCIs and FCIs (proving process omitted for the page limit).

C. Carbon flow tracing with circulating flows

In power system, circulating power exists due to the mal-operation control, the system complexity and poor controllability [13]. In fact, when there exists circulating flows in power system, the number of circulating nodes is usually very limited. Therefore, it is quite straight-forward to detect and find out the circulating part in the system while doing direct graph based tracing.

If the circulating part contains the sequence of nodes $(i_1, i_2, \dots, i_m, i_1)$, a circulating coefficient is defined for the proportion of power remaining after a circulation as:

$$T_i = \frac{P_{i_m i_1}}{P_{Ni_m}} \prod_{k=1}^{m-1} (P_{i_k i_{k+1}} / P_{Ni_k}) \quad (7)$$

Normally, if there exists a directed path between a pure generation node and a pure load node, denoted as $(j_1, j_2, \dots, j_n)$, the contribution effect along this path, namely $\prod_{k=1}^{n-1} (P_{j_k j_{k+1}} / P_{Nj_k})$, should be added to $D_{j_n j_1}$. If one of the nodes belongs to the circulation, this contribution effect to $D_{j_n j_1}$ will

be $\prod_{k=1}^{n-1} (P_{j_k j_{k+1}} / P_{Nj_k}) \cdot \sum_{d=0}^{+\infty} T_i^d$.

D. Carbon obligation allocation

When allocating carbon obligation at demand side, it is optional whether to include carbon losses. If only the load caused emission at each node is accounted

$$C_t = \sum_{k=1}^N NCI_k \times P_{Lk} + \sum_{i=1}^N \sum_{j \in \Gamma_+(i)} CL_{ij} \quad (8)$$

where C_t is the total carbon emission, P_{Lk} is the load at node k . Here the obligation allocated to consumers is the first part in equation (8), which is represented by the NCI. The extra obligation caused by carbon losses may then be obliged to the grid.

If the consumers are to be responsible for the carbon emission caused by network losses, the total carbon emission shall be expressed as

$$C_t = \sum_{k=1}^N FCI_k \times P_{Lk} \quad (9)$$

which is equivalent to transferring carbon losses to the end node of a line in the directed graph. Therefore, FCI increases along a single line in the system.

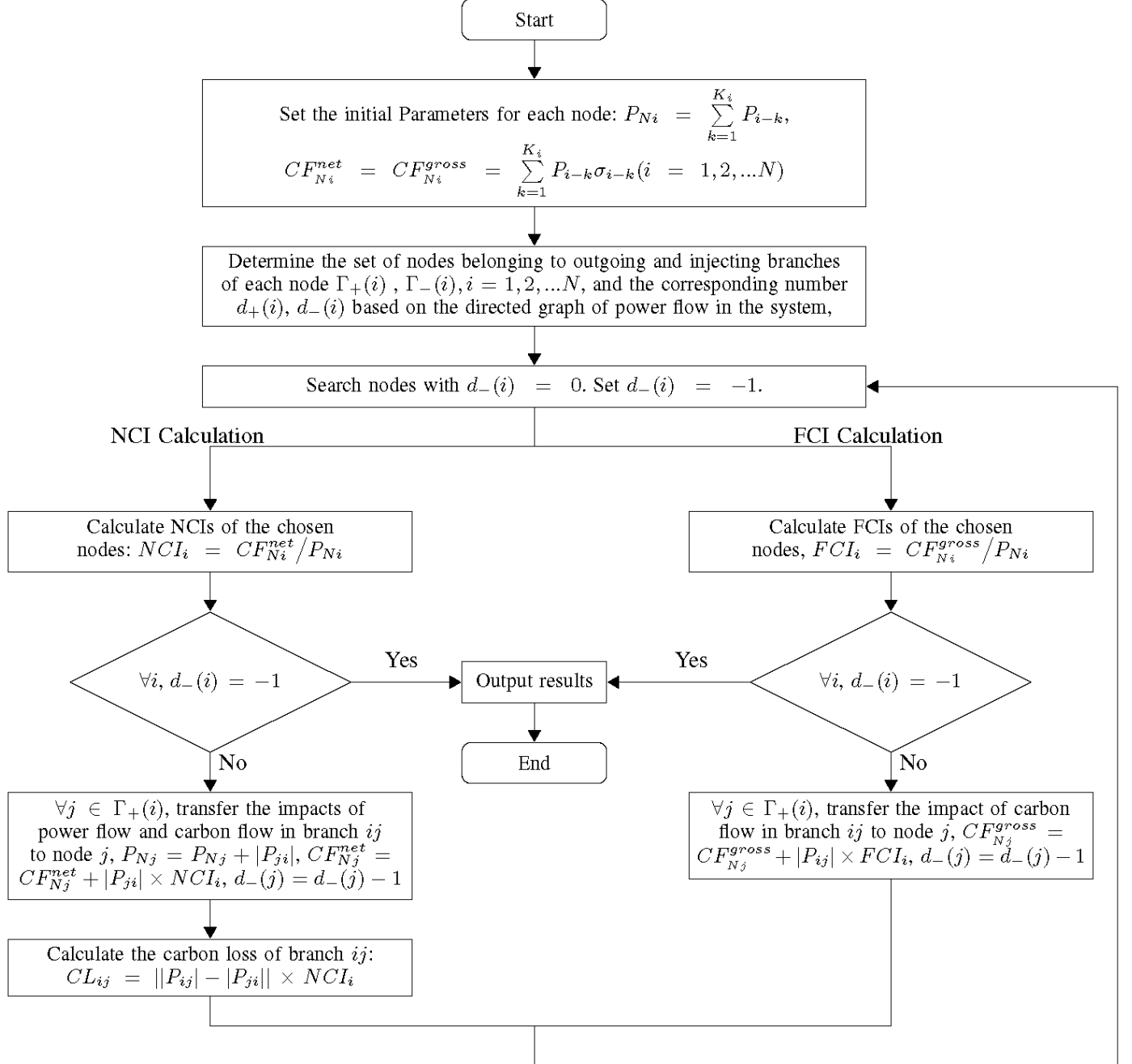


Fig. 1 Directed graph based method for tracing carbon flow in lossy networks

IV. CASE STUDY

For illustration purpose, we walk through the proposed method by presenting numerical results with data analysis of a distribution system and a large scale transmission system. All computations are performed on a 2.4-GHz PC with 8GB of RAM.

A. Distribution system case

The discussed distribution system, as a radial network, comprises 17 nodes and 16 lines. Various types of load are attached to some nodes including residential, industrial, commercial, municipal, traffic loads, etc. Fig. 2 shows the active power flow of the system, in which G1 and G2 represent a photovoltaic station and the grid infeed respectively. The directed graph based method is taken to calculate the carbon losses of distribution lines as well as the NCI and the FCI of

each node. The results can also be employed in fields such as generation expansion planning [14].

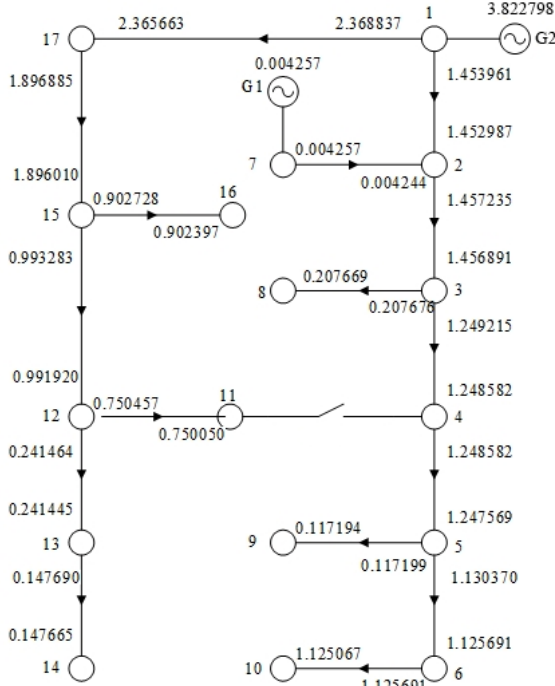


Fig. 2 Active power flow (MW) of the distribution network.

The carbon emission rates of generators in the mentioned system is 0.41436 tCO₂/MWh for G2 and 0.032 tCO₂/MWh for G1. The calculating process, as is described in Fig. 1 is then carried out based on the directed graph shown in Fig. 2. Two nodes (node 1 and 7) without injecting power flows are chosen as the initial points for solving. After all nodes in the directed graph have been eliminated, NCIs and FCIs of the nodes are obtained and shown in TABLE I. The carbon losses are obtained as shown in TABLE II.

TABLE I Carbon intensities of all nodes

Node #	NCI (tCO ₂ /MWh)	FCI (tCO ₂ /MWh)	FCI-NCI (tCO ₂ /MWh)
1	0.414360	0.414360	0.000000
2	0.413246	0.413524	0.000278
3	0.413246	0.413621	0.000375
4	0.413246	0.413831	0.000585
5	0.413246	0.414167	0.000921
6	0.413246	0.415889	0.002643
7	0.032000	0.032000	0.000000
8	0.413246	0.413630	0.000384
9	0.413246	0.414190	0.000944
10	0.413246	0.416119	0.002873
11	0.414360	0.415903	0.001543
12	0.414360	0.415677	0.001317
13	0.414360	0.415710	0.001350
14	0.414360	0.415780	0.001420
15	0.414360	0.415107	0.000747
16	0.414360	0.415260	0.000900
17	0.414360	0.414916	0.000556

TABLE II Power losses and carbon losses of all branches

Line #	From-To	Power Loss (MW)	Carbon Loss (tCO ₂ /h)
1	1 - 2	0.000974	0.000403
2	1 - 17	0.003174	0.001315
3	2 - 3	0.000344	0.000142
4	3 - 4	0.000633	0.000262

5	3 - 8	0.000007	0.000003
6	4 - 5	0.001013	0.000419
7	5 - 6	0.004679	0.001934
8	5 - 9	0.000005	0.000002
9	6 - 10	0.000624	0.000258
10	7 - 2	0.000013	0.000000
11	12 - 11	0.000407	0.000171
12	12 - 13	0.000019	0.000008
13	13 - 14	0.000025	0.000010
14	15 - 12	0.001363	0.000564
15	15 - 16	0.000331	0.000137
16	17 - 15	0.000875	0.000362

The FCI and the NCI data in TABLE I are visualized in two colored directed graphs shown in Fig. 3 and Fig. 4. The FCI is higher than or equal to the NCI at each node because the calculation of the FCI includes the effect of carbon losses. In fact, the FCIs increase along any directed path without other carbon sources breaking in. This growth effect along directed paths results that most high FCI nodes lie at the end of the network. On the other hand, the NCI distribution is divided into two parts (The NCIs for the right part and the left part are 0.414360 and 0.413246 tCO₂/MWh respectively) except node 7 (The NCI is 0.032000 tCO₂/MWh), which is the infeed of a photovoltaic station. The power generated by the photovoltaic station is all consumed in the left part. This is the reason for the separation of two parts. The carbon loss data in TABLE II is also visualized in Fig. 4. The carbon loss of a branch is affected not only by the power loss of the branch but also the NCI of its source node. For example, the line 10 (from node 7 to 2) has higher power loss than the line 5 (from node 3 to 8) and 8 (from node 5 to 9), but its carbon loss is lower.

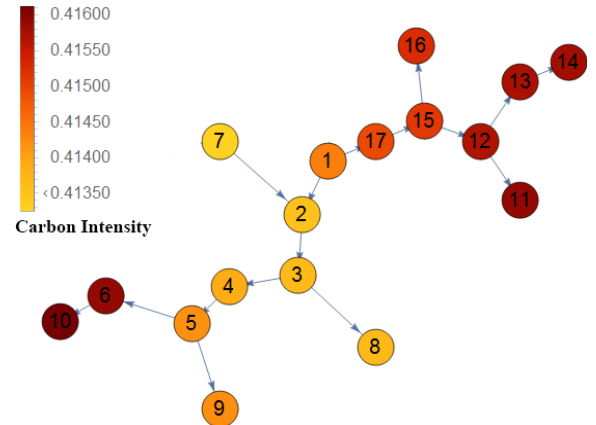


Fig. 3 FCI distribution in the network

B. Large scale transmission system case

A transmission system [15] of 2224 buses with more than 600 substations is studied here to apply the directed graph based carbon flow tracing method. In this case, about 1.7% of the lines have different flow directions on its two terminals as mentioned in Section III. The FCI distribution of the system is shown in Fig. 5, where the hollow circles mean substations without loads. This case is presented to show the effectiveness of the proposed method in a large scale power system. Further analysis is omitted here for the limited space.

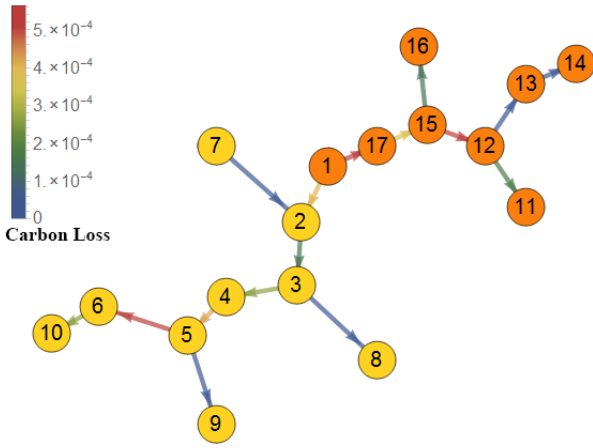


Fig. 4 NCI and carbon loss distribution in the network

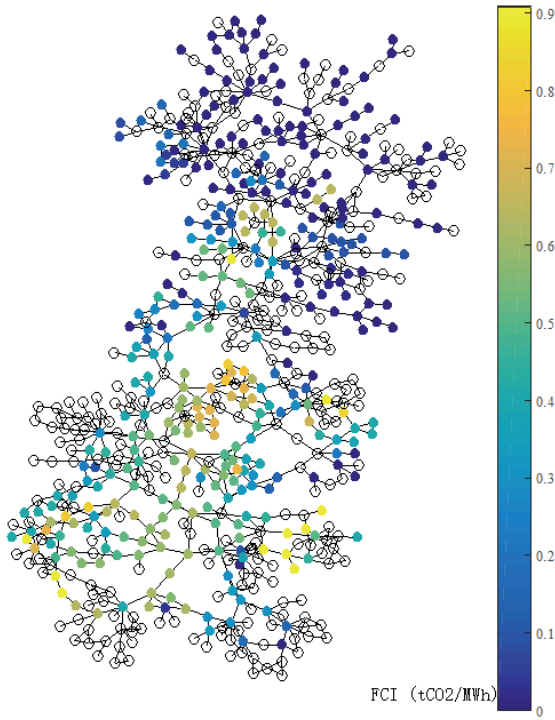


Fig. 5 FCI of the large scale power system

In order to evaluate the effectiveness of the proposed directed graph based method, the carbon flow tracing problem has been solved and compared to the case when the linear equation based method is applied (Table III). Note that directed graph based method requires much less time than the linear equation based method, which is more obvious as the system scale grows. Thus the proposed directed graph based carbon flow tracing method performs better for realistic large scale power systems than the existing linear equation based method.

Table III Comparison of solving carbon flow for different systems

Number of nodes	14	300	2224	3120	9241
Directed graph based	0.001s	0.011s	0.163s	0.236s	1.517s
Linear equation based	0.002s	0.194s	0.822s	2.402s	49.726s

V. CONCLUSIONS

A directed graph based method is proposed for carbon flow calculation in power systems. In particular, we present in the tracing process the calculation of carbon losses, FCIs and NCIs, which facilitates a direct carbon obligation allocation at demand side. The FCIs and NCIs are separately used to allocate carbon obligation to consumers with and without consideration of carbon losses. The directed graph based method avoids the matrix inverse calculation which may be time consuming. The

effectiveness of the proposed method is numerically tested in realistic distribution and transmission systems.

VI. ACKNOWLEDGEMENT

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